

3(a)(i)	pyrimidine ; <i>accept phonetic or close spellings, e.g. pyrimidine / pyrimidine / pirimidine / primidine</i>	1
3(a)(ii)	any one from: each nucleic acid has 4 (types of) base ; A has 4 bases DNA has A, T, C and G and RNA has A, U, C and G ; A full names even if misspelt unless another molecule (e.g. thiamine) <i>idea that</i> in RNA U replaces T / DNA has T but RNA has U ;	1
3(b)	any four from: <i>max 2 if answer is about transcription and mRNA</i> <i>penalise polypeptide once</i> 1 (triphosphate) activated / free, nucleotide (with thymine) ; 2 thymine / T, pairs with, adenine / A ; I complementary base pairing unqualified 3 adenine / A, on <u>template</u> strand ; 4 formation of two hydrogen bonds between T and A ; I H bonds between C and G 5 <i>ref to</i> DNA polymerase in correct context (e.g. correct base pairing / bond formation / proof reading) ; 6 forms phosphodiester bond (between -OH of deoxyribose and phosphate of nucleotide) ; R if DNA ligase 7 release of, pyrophosphate / P-P / two phosphates ; 8 nucleotide added to the 3' end of, newly-synthesised strand / leading strand / daughter strand (of DNA) ; A extension is in the 5' to 3' direction	4
3(c)	1 strands (are parallel and) run in <u>opposite</u> directions / one strand in 5' to 3' direction and other strand in 3' to 5' direction ; any two from: 2 elongation / synthesis / extension / AW, of strand is, in 5' to 3' direction / not in 3' to 5' direction ; 3 DNA polymerase, moves in the 5' to 3' direction / can only add nucleotides to the 3' end ; 4 <i>idea that</i> phosphate of each nucleotide is added to C3 of last nucleotide of growing strand ; 5 leading strand is made continuously / lagging strand is made up of Okazaki fragments ; 6 Okazaki fragments are attached together by DNA ligase ;	3